

Python Programming – Complete Exam Guide

PART A – 2 Marks

1. Output of $x=3/2$, $y=3//2$

Answer: 1

```
x=3/2
y=3//2
print(type(x),y)
```

2. Output of swapping

Answer: 10 15

```
a=5
b=10
a, b=b, a+b
print(a,b)
```

3. Literal collections

Answer: List, Tuple, Set, Dictionary

4. break vs continue

Answer: break stops loop, continue skips iteration

5. Split string

Answer: split() function

6. Dictionary output

Answer: {'a':5,'b':2}

```
d={'a':1, 'b':2}
d['a']=d.get('c',0)+5
print(d)
```

7. Count characters

```
for ch in s:
    d[ch]=d.get(ch,0)+1
```

8. Files vs modules

Answer: File stores data, module has functions/classes

9. Create file

```
open('example.txt', 'w').write('Hello')
```

10. NumPy output

Answer: [3 4 5]

```
a=np.array([1,2,3])  
print(a+2)
```

11. Pandas Series

Answer: Series is 1D, DataFrame is 2D

12. Swap variables

```
a, b=b, a
```

13. Recursion output

Answer: 6

```
def test(n): return 1 if n==0 else n*test(n-1)  
print(test(3))
```

14. Lambda add

```
lambda a,b:a+b
```

15. List output

Answer: [1,2,3,1,2,3]

```
L=[1, 2, 3]  
print(L*2)
```

16. Count tuple

```
t.count(2)
```

17. __init__ role

Answer: Constructor method

18. Custom exception

```
class NegativeNumberError(Exception): pass
```

19. Reshape array

```
a.reshape(1,2,3)
```

20. Pandas series

```
pd.Series([10,20,30],index=['a','b','c'])
```

PART B – 13 Marks

Arithmetic operations

```
print(a+b,a-b,a*b,a/b,a//b,a%b,a**b)
```

Employee details

```
print(name,age,address,designation,salary)
```

Operators

Answer: Arithmetic, Relational, Logical, Assignment, Bitwise

Loop with break & continue

```
for i in range(1,21):  
    if i>15: break  
    if i%5==0: continue  
    print(i)
```

Square root

```
import math  
math.sqrt(x)
```

Vowels & consonants

```
if ch in 'aeiou': v+=1 else: c+=1
```

Loan eligibility

```
if 25<=age<=50 and income>30000: print('Eligible')
```

Selection sort

```
arr.sort()
```

Histogram

```
d[ch]=d.get(ch,0)+1
```

Remove duplicates

```
list(set(l))
```

Invert dict

```
{v:k for k,v in d.items()}}
```

File word count

```
len(f.read().split())
```

Copy file

```
f2.write(f1.read())
```

Class Student

```
class Student: pass
```

Exception handling

```
try/except
```

Pandas group

```
df.groupby('class')
```

CSV operations

```
df.head(), df.shape
```

NumPy axis

```
a[:,None,:]
```

Armstrong

```
sum of cubes == number
```

Fibonacci

```
a,b=b,a+b
```

Simple interest

```
(p*r*t)/100
```

Factorial

```
loop multiplication
```

PART B – 15 Marks

Sales dictionary

```
d=dict(sales)
```

Above 100

```
{k:v for k,v in d.items() if v>100}
```

Swap max & min

```
max_key=min_key swap
```

Histogram

```
print('*'*v)
```

Student dictionary list

```
students=[{}]
```

List methods

Answer: append(), remove(), sort()